

### EXPERIMENT NO.3

**AIM- MEASUREMENT OF POWER IN THREE PHASE CIRCUIT BY TWO WATTMETER METHOD**

Measurement of Power in three phase circuit by Two Wattmeter method apparatus is a versatile & control panel to study the following experiment.

**Experiment No. 1:-** Measurement of apparent power in three phase circuit by two wattmeter method.

**Experiment No. 2:-** Measurement of real power (Active & Reactive Power)

#### INSTRUMENTS REQUIRED:

To conduct the above mentions experiment we require three phase electrical connector 1 No. resistive amp load 1.2KW & 1No. of three phase inductive load 6 Amp., patch cord & instruction manual.

The Control panel consists of the following Built in parts :-

1. Three Nos. of Moving Coil Voltmeter of Range 500VAC of size 96\*96mm provided with Input Terminals.
2. Three Nos. of Moving Coil Ammeter of Range 3AAC of size 96\*96mm provided with Input Terminals.
3. Two Nos. of Single Phase Wattmeter of Range 500W of size 96\*96mm provided with Input Terminals.
4. One No. of Power Factor Meter of size 96\*96mm provided with Input Terminals.
5. One No. of Miniature Circuit Breaker of Range 415V/ 10 Amps (MCB/ TPN) Provided on the Input Side.

6. Circuit Diagram printed on Bakelite Sheet from Connecting Terminals.

7. Housed in wooden box is in a Tapered 'shape'

8. Dimension (mm): 470(L) x 460(B) 620(H)

9. Power Requirement: Three Phase 415VAC.

#### THEORY:

In a three phase three wire system: we require three the common points of the pressure coils coincide with will require only  $n - 1 = 2$  elements.

Instantaneous power consumed by load =  $V_1 I_1 + V_2 I_2 + V_3 I_3$

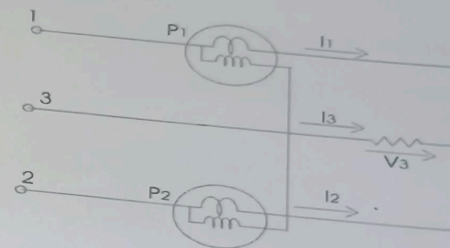


FIG. (1) TWO WATT METER METHOD

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6. Circuit Diagram printed on Bakelite Sheet front panel With Instruments Connecting Terminals.
7. Housed in wooden box is in a Tapered 's' shape for better view angle.
8. Dimension (mm): 470(L) x 460(B) 620(H)
9. Power Requirement: Three Phase 415VAC.

#### THEORY:

In a three phase three wire system: we require three elements, but if we make the common points of the pressure coils coincide with one of the lines, then we will require only  $n - 1 = 2$  elements.

Instantaneous power consumed by load =  $V_1 I_1 + V_2 I_2 + V_3 I_3$

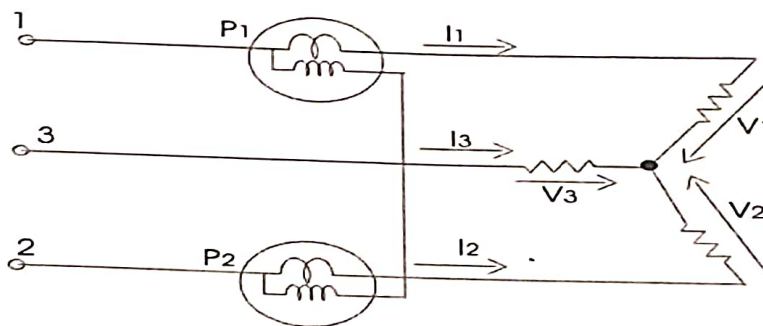


FIG. (1) TWO WATT METER METHOD(STAR CONNECTION)



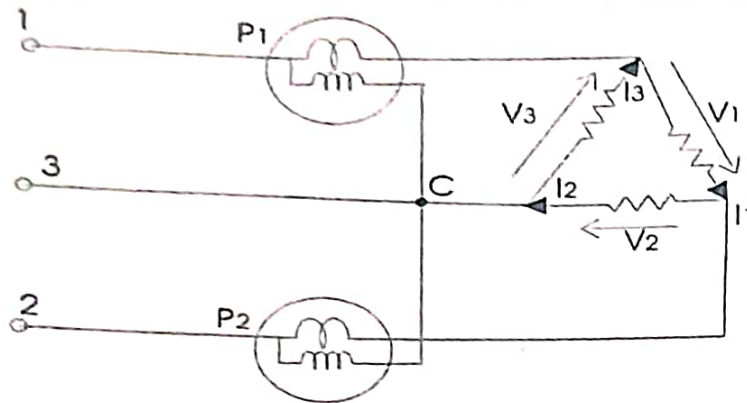


FIG. (2) TWO WATT METER METHOD (DELTA CONNECTION)

Let us consider two wattmeter connected to measure power in three phase circuit as shown in Fig.(1) as Star Connection and Fig. (2) as Delta Connection.

**Star Connections:**

Instantaneous reading of  $P_1$  wattmeter,  $P_1 = I_1 (V_1 - V_3)$ .

Instantaneous reading of  $P_2$  wattmeter,  $P_2 = I_2 (V_2 - V_3)$ .

Sum of instantaneous reading of two wattmeter's =  $P_1 + P_2$

$$= I_1 (V_1 - V_3) + I_2 (V_2 - V_3).$$

From Kirchhoff's Law as shown in Fig. (1)

$$I_1 + I_2 + I_3 = 0$$

Or

$$I_3 = -(I_1 + I_2)$$

Therefore, the sum of instantaneous readings of two wattmeter's

$$= V_1 I_1 + V_2 I_2 + V_3 I_3$$

Therefore, the sum of the two-wattmeter reading is equal to the power consumed by the load. This is irrespective of whether the load is balanced or unbalanced.

### Delta Connection:

Instantaneous reading of  $P_1$  wattmeter,  $P_1 = -V_3 (I_1 - I_2)$

Instantaneous reading of  $P_2$  wattmeter,  $P_2 = V_2 (I_2 - I_1)$

Therefore Sum of instantaneous reading of watt meters  $P_1$  &  $P_2$ :

$$\begin{aligned} P_1 + P_2 &= -V_3 (I_1 - I_2) + V_2 (I_2 - I_1) \\ &= V_2 I_2 + V_3 I_3 - I_1 (V_2 + V_3) \end{aligned}$$

Current through wattmeter  $P_2$  is  $I$

and voltage across its pressure coil is  $V_{23}$ .  $I_2$  lags  $V_{23}$  by an angle.

Reading of  $P_2$  wattmeter

$$\begin{aligned} P_2 &= V_{23} I_2 \cos (30^\circ + \phi) \\ &= \sqrt{3} VI \cos (30^\circ + \phi) \end{aligned}$$

Sum of reading of two wattmeter's:

$$\begin{aligned} P_1 + P_2 &= \sqrt{3} VI [\cos (30^\circ - \phi) - \cos (30^\circ + \phi)] \\ &= 3 VI \cos \phi \end{aligned}$$

This is the total power consumed by load.

Therefore, total power consumed by load:

$$P = P_1 + P_2$$

Difference of readings of two wattmeter's:

$$\begin{aligned} P_1 - P_2 &= \sqrt{3} VI [\cos (30^\circ - \phi) - \cos (30^\circ + \phi)] \\ &= \sqrt{3} VI \sin \phi \end{aligned}$$

Therefore,

$$\frac{P_1 - P_2}{P_1 + P_2} = \frac{\sqrt{3} VI \sin \phi}{3 VI \cos \phi} = \frac{\tan \phi}{\sqrt{3}}$$

$$\phi = \tan^{-1} \sqrt{3} \frac{P_1 - P_2}{P_1 + P_2}$$

Power Factor

$$\cos \phi = \cos \tan^{-1} \sqrt{3} \frac{P_1 - P_2}{P_1 + P_2}$$

Effect of power factor on the readings of wattmeter's:

With unity power factor  $\cos = 1$  and  $\phi = 0$ .

The readings of two wattmeter's are:

$$P_1 = \sqrt{3} VI \cos (30^\circ - \phi)$$

$$= \sqrt{3} VI \cos 30^\circ$$

$$= (3/2) VI$$

$$P_2 = \sqrt{3} VI \cos (30^\circ - \phi)$$

$$= \sqrt{3} VI \cos 30^\circ$$

$$= (3/2) VI$$

$$P_1 + P_2 = 3 VI$$

At unity power factor, total power =  $P = 3 VI \cos \phi$  &  $= 3 VI$

Thus, at unity power factor, the readings of the two wattmeter's are equal, each wattmeter reads half of total power.

When

$$\text{P.F} = 0.5$$

$$\phi = 60^\circ$$

Therefore,

$$P_1 = \sqrt{3} VI \cos (30^\circ - \phi)$$

$$= \sqrt{3} VI \cos (30^\circ - 60^\circ)$$

$$= (3/2) VI$$

$$P_2 = \sqrt{3} VI \cos (30^\circ + \phi)$$

$$= \sqrt{3} VI \cos (30^\circ + 60^\circ)$$

$$= 0$$

$$P_1 + P_2 = (3/2) VI + 0$$

$$= (3/2) VI$$

Total power

$$P = 3 VI \cos = (3/2) VI$$

Therefore, when the power factor is 0.5, one of the wattmeter's reads zero and the other reads total power

When

$$\text{P.F} = 0$$

Have

$$\phi = 90^\circ$$

Therefore

$$P_1 = \sqrt{3} VI \cos (30^\circ - \phi)$$

$$= \sqrt{3} VI \cos (30^\circ - 90^\circ)$$

$$= (3/2) VI$$

$$P_2 = \sqrt{3} VI \cos (30^\circ + \phi)$$

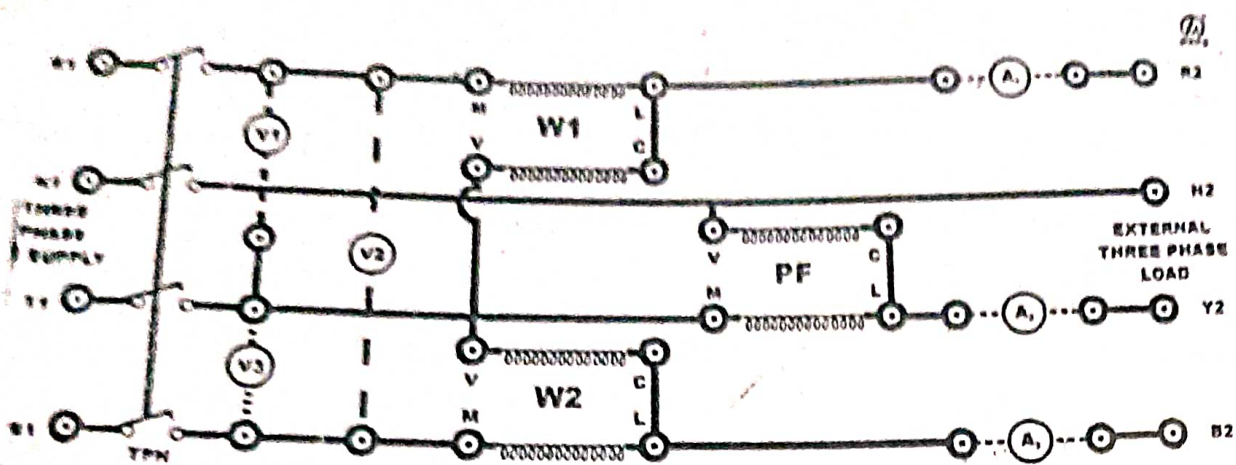


$$= \sqrt{3} V I \cos (30^\circ + 90^\circ)$$

$$= -[\sqrt{3}/2] V I$$

$$P_1 + P_2 = 0.$$

Power  $P = 3 V I \cos = 0$



Therefore, with zero power factor, the readings of the two wattmeters are equal but of opposite sign negative. It should be noted that when the power factor is below 0.5, one of the wattmeter will give indication. under these conditions in order to read the wattmeter, we must either reverse the current coil or the pressure coil connections. The wattmeter will then give a positive reading but this must be taken as negative for calculating the total power.



## PROCEDURE

**Experiment No. 1:-** Measurement of power in three phase circuit using two wattmeter method & calculation of apparent power.

- Connect the meters with circuit on panel as per diagram with the help of connecting wires.
- Connect the external three phase supply to the left hand side of the panel marked R1, Y1, B1 & N1.
- Connect the external three phase resistive load 1.2KW arrangement to the terminals provided at the right hand side of the panel marked R2, Y2, B2 & N2.
- Ensure that load is at minimum or OFF state.
- Switch ON the supply and measure the Line Voltage across three phase supply.
- Vary the load in each phase, so that equal readings are obtained in three ammeter & in three voltmeters.
- Note down the readings of wattmeter, voltmeters & ammeters.
- Calculate the total power consumed from these readings and verify the method.

Total power consumed is equal to the sum of two wattmeter's i.e.

$$P = W_1 + W_2$$

Calculate the power factor of the circuit by using the formula:

$$\cos \phi = \cos \tan^{-1} \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2}$$

Repeat the process and take different readings for calculation purpose.

**OBSERVATION TABLE No. :- 1**

[illegible]

**Experiment No. 2:-** Measurement of power factor, active & reactive power

**Procedure :-**1 Repeat steps four to ten by connecting inductive load in stand of resistive lamp load & observation in table No. 2. at the process & take different reading for calculation purpose.

**OBSERVATION TABLE No. :- 2**

| S.No. | OBSERVATION     |                 |                 |                |                |                |               |               | CALCULATIONS     |                                     |                                                           |
|-------|-----------------|-----------------|-----------------|----------------|----------------|----------------|---------------|---------------|------------------|-------------------------------------|-----------------------------------------------------------|
|       | Phase Voltage   |                 |                 | Phase Current  |                |                | Watts         |               | Measured         | Calculated                          | $\cos \phi = \cos \tan^{-1} \sqrt{3} \frac{W1-W2}{W1+W2}$ |
|       | Vph1<br>(Volts) | Vph2<br>(Volts) | Vph3<br>(Volts) | Iph1<br>(Amps) | Iph1<br>(Amps) | Iph1<br>(Amps) | W1<br>(Watts) | W2<br>(Watts) | W1+W2<br>(Watts) | $\sqrt{3} V_l \cos \phi$<br>(Watts) |                                                           |
|       |                 |                 |                 |                |                |                |               |               |                  |                                     |                                                           |
|       |                 |                 |                 |                |                |                |               |               |                  |                                     |                                                           |
|       |                 |                 |                 |                |                |                |               |               |                  |                                     |                                                           |
|       |                 |                 |                 |                |                |                |               |               |                  |                                     |                                                           |
|       |                 |                 |                 |                |                |                |               |               |                  |                                     |                                                           |
|       |                 |                 |                 |                |                |                |               |               |                  |                                     |                                                           |
|       |                 |                 |                 |                |                |                |               |               |                  |                                     |                                                           |
|       |                 |                 |                 |                |                |                |               |               |                  |                                     |                                                           |

### PRECAUTIONS

1. The current should not exceed the rated value.
2. Don't touch naked wire.
3. Check for the proper connections with the optional accessories.
4. Do not short the phase to phase or phase to neutral connections.



### Questions

1. What is a two-wattmeter method?
2. What are the key components of a typical wattmeter?

**Current Coils:**

**Potential Coils:**

3. If the reading of the two wattmeter's is equal and opposite while measuring power in a 3-phase induction motor then the power factor of the load will be?
4. In a 3 phase power measurement by two wattmeter method, both the wattmeter's had identical readings. The power factor of the load was?
5. Two wattmeter method is used to measure three-phase \_\_\_\_ load.
6. The wattmeter method is used to measure power in a three-phase load. The wattmeter readings are 400W and -35W.  
Calculate the total active power  
Find the power factor.  
Find the reactive power.